

1. Administrative & Study Identification

Official Study Title: A Phase 1/2 Open-Label Rolling-Arm Umbrella Platform Study of Investigational Agents With or Without Pembrolizumab in Participants with PD-1/L1 Refractory Locally Advanced or Metastatic Urothelial Carcinoma (KEYMAKER-U04): Substudy 04A **Study Title:** A Substudy of Investigational Agents in Programmed Cell Death-1/Ligand 1 (PD-1/L1) Refractory Locally Advanced or Metastatic Urothelial Carcinoma (mUC) (MK-3475-04A) **Short Title/Acronym:** KEYMAKER-U04 Substudy 04A (MK-3475-04A) **Protocol Number:** MK-3475-04A **NCT Number:** NCT05562830 **Posting ID:** 536567089285585348 **Sponsor/Company Name:** Merck / Merck Sharp & Dohme LLC **Investigational Product:** Investigational agents (e.g., Zilovertamab vedotin / MK-2140) with or without Pembrolizumab (Trade Name: KEYTRUDA®, ATC Code: L01FF02) **Phase of Development:** PHASE1, PHASE2 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Merck Clinical Trial Representative / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and tolerability of the study interventions by evaluating the percentage of participants who experience at least one adverse event (AE) and those who discontinue treatment due to an AE, and to evaluate the Objective Response Rate (ORR) in Arm A as assessed by Blinded Independent Central Review (BICR). **Secondary Objectives:** To evaluate the duration of response (DOR) for Arm A as assessed by BICR, and for Arm B as assessed by the Investigator.

2.2 Background & Rationale Locally advanced or metastatic urothelial carcinoma (mUC) is a serious and incurable condition with poor long-term survival and a high unmet medical need. This trial serves as a substudy (04A) of an umbrella platform (KEYMAKER-U04) designed to evaluate new investigational agents (with or without pembrolizumab) for participants whose disease is resistant/refractory to standard programmed cell death-1/ligand 1 (PD-1/L1) inhibitors.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Platform Study) **Control Type:** Active/Rolling-Arm (Participants assigned to rolling investigational treatment arms) **Blinding:**

Open-label **Randomization:** N/A (Rolling-arm assignment based on protocol infrastructure)

3.2 Planned Enrollment & Duration Targeted Enrollment: 48 participants **Number of Sites:** Multicenter **Estimated Study Start:** 16-Nov-2022 **Estimated Primary Completion:** 28-Oct-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years old. 2. Histologically or cytologically confirmed diagnosis of locally advanced/unresectable or mUC of the renal pelvis, ureter (upper urinary tract), bladder, or urethra. 3. **Arm A:** PD-1/L1 refractory locally advanced or mUC as evidenced by EITHER disease progression while on treatment or after treatment with an anti-PD-1/L1 monoclonal antibody (mAb) for locally advanced/unresectable or mUC, OR disease recurrence while on treatment or after treatment with an anti-PD-1/L1 mAb for muscle-invasive urothelial carcinoma (MIUC). Participants must provide an archival tumor tissue sample or newly obtained core or excisional biopsy of a tumor lesion demonstrating UC, not previously irradiated, and adequate for biomarker evaluation. 4. **Arm B:** PD-1/L1 refractory locally advanced or mUC as evidenced by EITHER disease progression after treatment with an anti-PD-1/L1 mAb for locally advanced/unresectable or mUC, OR disease recurrence after treatment with an anti-PD-1/L1 mAb for MIUC administered as monotherapy or in combination with other checkpoint therapies >12 months after last dose. Participants must provide an archival tumor tissue sample or newly obtained core or excisional biopsy of a tumor lesion from a metastatic site or from a primary tumor that has become locally advanced and not previously irradiated.

4.2 Exclusion Criteria 1. Known additional nonurothelial malignancy that is progressing or has required active treatment within 3 years prior to study randomization/allocation. 2. Received prior systemic anticancer therapy, including investigational agents, within 4 weeks before randomization/allocation. 3. Active infection requiring systemic therapy. 4. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines is allowed. 5. Known history of human immunodeficiency virus (HIV). 6. Known history of hepatitis B or known hepatitis C virus infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab (KEYTRUDA®) administered every 3 weeks or every 6 weeks. Investigational treatments (e.g., Zilovetamab vedotin) administered based on specific arm assignment. **Route of Administration:** Intravenous (IV). **Duration of Treatment:**

Approximately 2 years (treatment may continue as long as the cancer does not worsen and drugs are tolerated).

5.2 Concomitant & Prohibited Therapy Permitted: Administration of killed vaccines and standard supportive care regimens.

Prohibited: Live or live-attenuated vaccines within 30 days prior to the first dose; prior systemic anticancer therapies/investigational agents within 4 weeks prior to allocation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
:---	:---	:---
:---	:---	Screening ~1 Month Informed Consent, Physical Examinations, ECGs, Blood/Urine Samples, Scans, Tissue Biopsy
	Baseline Day 0	Rolling-arm assignment, First Dose
	Interim Every 3 or 6 Weeks	Study Drug Administration (e.g., pembrolizumab), Safety Labs, AEs Assessment
	End of Study Up to 5 Years	Final Scans, Survival Follow-Up, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Percentage of Participants Who Experienced At Least One Adverse Event (AE):** Defined as any unfavorable and unintended sign, symptom, disease, or worsening of preexisting condition temporally associated with study treatment and irrespective of causality. 2. **Percentage of Participants Who Discontinued Study Treatment Due to an AE.** 3. **Arm A: Objective Response Rate (ORR) as Assessed by Blinded Independent Central Review (BICR):** ORR is defined as the percentage of participants who achieve a Complete Response (CR) or Partial Response (PR) per RECIST 1.1.

7.2 Secondary & Exploratory Endpoints 1. **Arm A:** Duration of Response (DOR) as Assessed by BICR. 2. **Arm B:** Duration of Response (DOR) as Assessed by Investigator.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via physical examinations, ECGs, blood/urine samples, and clinical assessments throughout the treatment phase. **Serious Adverse Events (SAEs):** Reported promptly following standard Phase 1/2 oncology protocol guidelines. **Data Monitoring Committee (DMC):** Internal sponsor review committees oversee safety in the umbrella platform.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for efficacy outcomes (e.g., ORR, DOR); Safety Set for AE evaluations. **Sample Size Justification:** Total targeted enrollment of 48 participants across treatment arms is considered sufficient for early-phase estimation of safety, tolerability, and preliminary ORR. **Handling of Missing Data:** Handled as per standard oncology rules (e.g., non-responders or right-censoring for time-to-event outcomes).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of trial sites to ensure protocol adherence. **Audit Trail:** Continuous eCRF locking, source data verification (SDV), and data clarification via sponsor's quality systems.

11. Study Identifiers & Governance

Primary ID: MK-3475-04A **Registry Number:** NCT05562830 **Posting ID:** 536567089285585348 **Sponsor/Lead Organization:** Merck / Merck Sharp & Dohme LLC **Study Title:** KEYMAKER-U04 Substudy 04A

12. Clinical Framework (Oncology Specific)

Disease Area: Urothelial Carcinoma **Preferred UMLS Name:** Refractory Metastatic Urothelial Carcinoma **Primary Condition:** Locally Advanced or Metastatic Urothelial Carcinoma (mUC) **Diagnosis Threshold:** PD-1/L1 refractory (disease progression during or after anti-PD-1/L1 mAb therapy or recurrence after treatment for MIUC) **Medical Methodology:** Experimental investigational agents $\pm$ Pembrolizumab **Optimization Target:** Safety profile, Objective Response (ORR), and Duration of Response (DOR)

13. Study Procedures & Methodology

Management Pathway: Specialized Phase 1/2 Umbrella oncology protocol **Education Protocol (HCP):** Investigator training on complex rolling-arm umbrella design assignment **Education Protocol (Patient):** Informed consent detailing investigational assignments and treatment risks **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for impartial efficacy endpoint assessment

14. Observation & Follow-Up Schedule

Total Duration: ~5 years (from first participant consent to last trial-related contact) **Onsite Visit Cadence:** Every 3 or 6 weeks during the ~2-year treatment phase **Remote Monitoring Frequency:** Continuous throughout the follow-up survival period. **Checklist Review Interval:** Scans and assessments per standard RECIST 1.1 schedules.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					